# Permutations and Combinations

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## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Permutations & Combinations Practice Exercises

Find the number of ways to arrange the 7 letters in the word SUCCESS.

**2** [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many ways can 5 people be seated in a row if two particular people (A and B) must not sit next to each other?

**3** [Foundational] • Target: Permutations & Combinations Practice Exercises

The circle $C$ has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. (a) Find the coordinates of the centre and the exact radius of $C$. (b) Find the equation of the tangent to $C$ at the point $P(6, 2)$.

**4** [Foundational] • Target: Permutations & Combinations Practice Exercises

Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle $\theta$ between them to the nearest tenth of a degree.

**5** [Foundational] • Target: Permutations & Combinations Practice Exercises

Find the equation of the normal to the curve $y = \frac{4}{\sqrt{x}} + 2x$ at the point where $x = 4$.

**6** [Foundational] • Target: Permutations & Combinations Practice Exercises

Solve the equation $2\cos 2x = 1 + \sin x$ for $0 \le x \le 2\pi$ radians, giving answers to 3 significant figures.

**7** [Foundational] • Target: Permutations & Combinations Practice Exercises

A block of mass $8\ \text{kg}$ rests in limiting equilibrium on a rough plane inclined at $30^\circ$ to the horizontal. Taking $g = 9.8\ \text{m/s}^2$: (a) calculate the normal reaction $R$, (b) find the coefficient of friction $\mu$ between the block and the plane.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Permutations & Combinations Practice Exercises

Find the constant term (independent of $x$) in the binomial expansion of $\left(2x^2 - \frac{1}{x}\right)^9$.

**9** [Intermediate] • Target: Permutations & Combinations Practice Exercises

The curve $C$ has equation $x^3 - 2xy + y^2 = 4$. Find the coordinates of the two points on $C$ where the tangent is parallel to the y-axis (vertical).

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**10** [Intermediate] • Target: Permutations & Combinations Practice Exercises

A bag contains 5 red and 4 green marbles. Three marbles are drawn at random without replacement. Find the probability that: (a) all three are red, (b) at least two are red.

**11** [Intermediate] • Target: Permutations & Combinations Practice Exercises

A train accelerates uniformly from rest to $30$ m/s in $20$ s, travels at constant speed $30$ m/s for $60$ s, and then decelerates uniformly to rest in $15$ s. Find: (a) the acceleration in the first phase, (b) the total distance travelled, (c) the average speed for the entire journey.

**12** [Intermediate] • Target: Permutations & Combinations Practice Exercises

Convert the Cartesian equation $(x^2 + y^2)^2 = 4(x^2 - y^2)$ into polar form and identify the name and key features of the resulting curve.

**13** [Intermediate] • Target: Permutations & Combinations Practice Exercises

A block of mass $4$ kg is pulled along a rough horizontal plane ($\mu = 0.25$) by a pulling force of magnitude $25$ N acting at $30^\circ$ above the horizontal. Taking $g = 9.8$ m/s$^2$: (a) show that the normal reaction $R = 26.7$ N, (b) find the acceleration of the block.

**14** [Intermediate] • Target: Permutations & Combinations Practice Exercises

Two particles $P$ of mass $3$ kg and $Q$ of mass $5$ kg are connected by a light inextensible string passing over a smooth fixed pulley. The system is held at rest and released. Taking $g = 9.8$ m/s$^2$: (a) find the acceleration $a$ of the system, (b) calculate the tension $T$ in the string, (c) find the force exerted by the string on the pulley.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

A continuous random variable $X$ has PDF given by $f(x) = \frac{k}{x^2}$ for $1 \le x \le 4$, and $0$ elsewhere. (a) Show that $k = \frac{4}{3}$. (b) Find the median of $X$.

**16** [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

Decompose the rational expression $R(x) = \frac{7x - 1}{x^2 - x - 6}$ into partial fractions using the Heaviside cover-up method.

**17** [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

Write the standard equation of the ellipse having foci at $(\pm 3, 0)$ and vertices at $(\pm 5, 0)$. State the lengths of the major and minor axes, and compute the eccentricity.

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**18** [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

**19** [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

The number of accidents on a highway follows a Poisson distribution with mean 1.6 per week. (a) Find the probability of at least 3 accidents in a 2-week period. (b) Given that at least one accident occurs in a week, find the probability that there are exactly 2 accidents in that week.

**20** [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

Find the equation of the perpendicular bisector of the line segment joining $P(2, -1)$ and $Q(8, 7)$. Give your answer in the form $y = mx + c$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Permutations & Combinations Practice Exercises

Evaluate the number of permutations \( {}^{5}P_{2} \).

**a.** 25

**b.** 21

**c.** 10

**d.** 20

**22** [Foundational] • Target: Permutations & Combinations Practice Exercises

A committee of 3 people is to be selected from a group of 5 people. The number of ways to form this committee is given by the combination formula ${}^nC_r = \frac{n!}{r!\,(n-r)!}$. How many different committees can be formed?

**a.** 10

**b.** 15

**c.** 20

**d.** 60

**23** [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many distinct ways can $6$ different people be seated around a circular table (rotations considered identical)?

**a.** 120

**b.** 103

**c.** 122

**d.** 100

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**24** [Foundational] • Target: Permutations & Combinations Practice Exercises

Find the number of distinct arrangements of the letters of the word PUPPY.

**a.** 103

**b.** 27

**c.** 64

**d.** 60

**25** [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many ways can 8 different books be arranged in a row on a shelf?

**a.** 45808

**b.** 35108

**c.** 40625

**d.** 40320

**26** [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many ways can 6 different people be seated in a row of 6 chairs?

**a.** 720

**b.** 817

**c.** 800

**d.** 725

**27** [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many ways can 11 different people be arranged in a row if two particular people must NOT sit next to each other?

**a.** 32806546

**b.** 33364823

**c.** 34615086

**d.** 32659200

**28** [Foundational] • Target: Permutations & Combinations Practice Exercises

In how many ways can 8 different people be arranged in a row if two particular people must sit together?

**a.** 10080

**b.** 8546

**c.** 9551

**d.** 9288

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** [Intermediate] • Target: Permutations & Combinations Practice Exercises

In how many ways can $6$ different runners be lined up at the start of a race if two particular runners must stand next to each other?

**a.** 273

**b.** 240

**c.** 220

**d.** 241

**30** [Intermediate] • Target: Permutations & Combinations Practice Exercises

In how many ways can $5$ of $7$ different people be selected and arranged in a row?

**a.** 2891

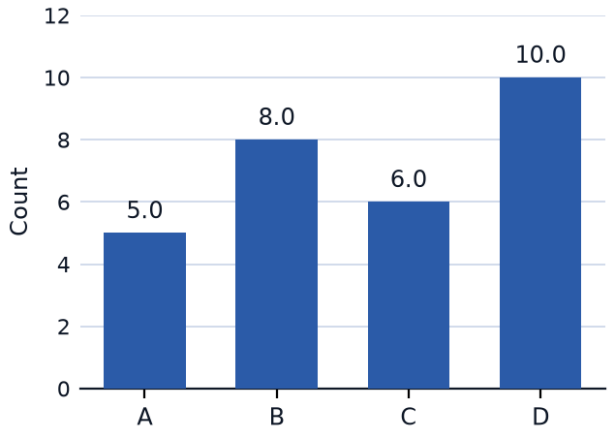
**b.** 2282

**c.** 2520

**d.** 2348

**31** [Intermediate] • Target: Permutations & Combinations Practice Exercises

The histogram below shows the distribution of heights of a group of students. The total number of students in the group is 50. If the frequency density for the class interval $60 \leq h < 70$ is $1.2$, find the value of $k$.




**a.** 10

**b.** 12

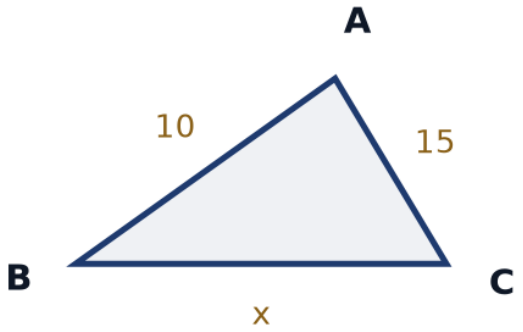
**c.** 15

**d.** 20

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**32** **[Intermediate]** • **Target: Permutations & Combinations Practice Exercises**

In triangle ABC, angle ABC = $\theta$ degrees, and angle ACB = $2\theta$ degrees. The lengths of sides AB and AC are 10 cm and 15 cm respectively. The length of BC is denoted by x cm. Given that the area of triangle ABC is 60 $cm^2$, find the value of x, to 3 significant figures.




**a.** 13.7

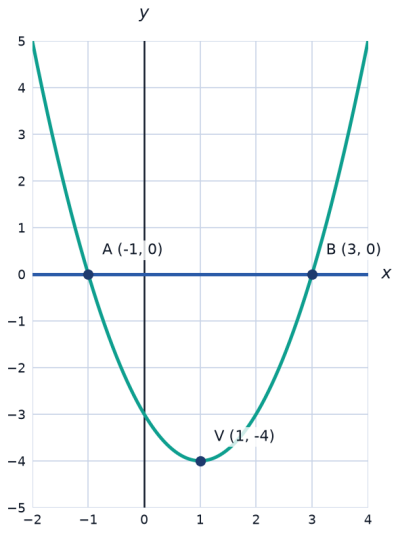
**b.** 11.4

**c.** 12.8

**d.** 14.9

**33** **[Intermediate]** • **Target: Permutations & Combinations Practice Exercises**

The diagram shows a curve with equation $y = ax^2 + bx + c$ and a straight line with equation $y = mx + d$. The curve and the line intersect at points A and B. The vertex of the parabola is at point V. The x-coordinates of the points A, B, and V are given as $x_A = -1$, $x_B = 3$, and $x_V = 1$ respectively. The y-coordinate of the vertex V is $y_V = -4$. Determine the value of $a$.




**a.** 1

**b.** 2

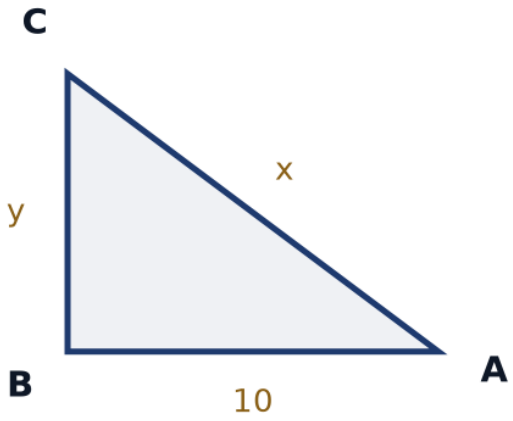
**c.** -1

**d.** -2

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**34** [Intermediate] • Target: Permutations & Combinations Practice Exercises

In triangle ABC, angle ABC = 90°, AB = 10 cm, and angle BAC = $\theta$. Given that $\sin\theta = \frac{3}{5}$, find the length of AC.




**a.** 16.7 cm

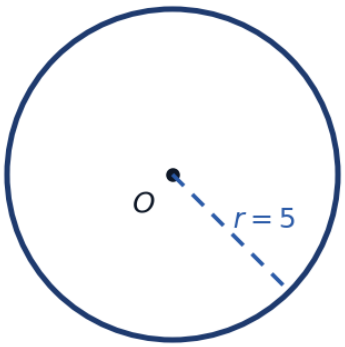
**b.** 8.0 cm

**c.** 12.5 cm

**d.** 6.0 cm

**35** [Intermediate] • Target: Permutations & Combinations Practice Exercises

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point $P$ is given by $3x + 4y = k$. Find the value of $k$.




**a.** 25

**b.** 30

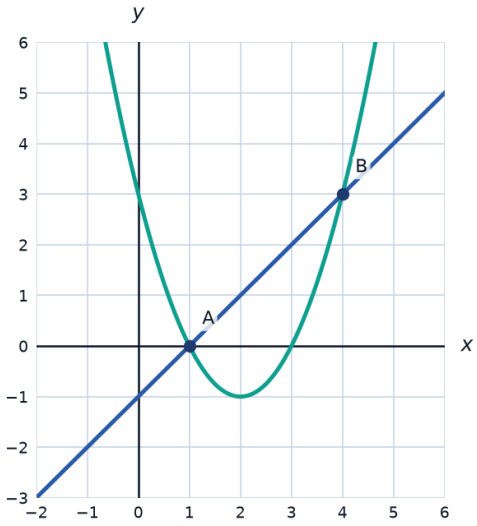
**c.** 35

**d.** 40

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**36** **[Intermediate]** • **Target: Permutations & Combinations Practice Exercises**

The graph of the quadratic function $y = x^2 - 4x + 3$ is shown. The graph of the linear function $y = x - 1$ is also shown. The graphs intersect at points A and B. Find the coordinates of point A.




**a.** (1, 0)

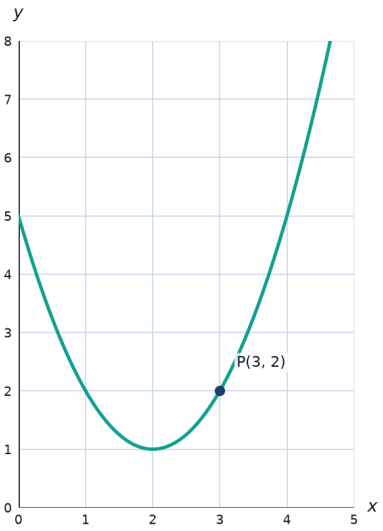
**b.** (3, 2)

**c.** (0, 3)

**d.** (2, 1)

**37** **[Intermediate]** • **Target: Permutations & Combinations Practice Exercises**

The diagram shows part of the curve with equation $y = x^2 - 4x + 5$ and the point $P(3, 2)$ lying on the curve. Find the gradient of the tangent to the curve at the point $P$.




**a.** 2

**b.** 6

**c.** $-4$

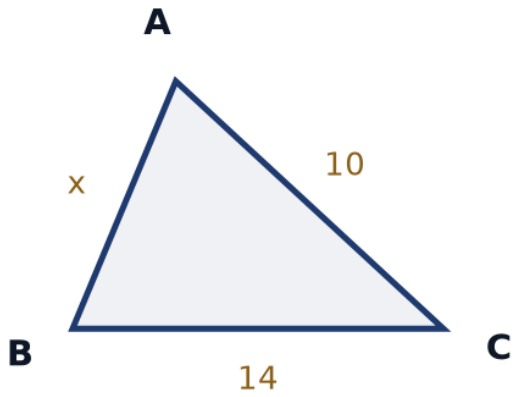
**d.** $\frac{2}{3}$

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** **[Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities**

In triangle ABC, angle A is 52 degrees. The side opposite angle B (AC) has a length of 10 units, and the side opposite angle C (AB) has a length of 14 units. What is the length of the side opposite angle A (BC)?




**a.** 10.3

**b.** 12.1

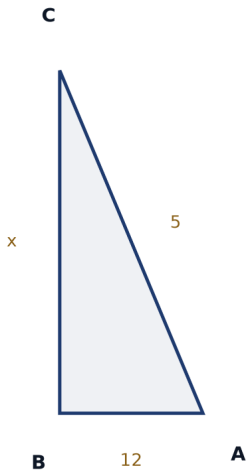
**c.** 14.8

**d.** 8.5

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**39** [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

In the right-angled triangle ABC, $\angle ABC = 90^\circ$. If $AB = 5$ cm and $BC = 12$ cm, find the length of the hypotenuse AC.




**a.** 10 cm

**b.** 13 cm

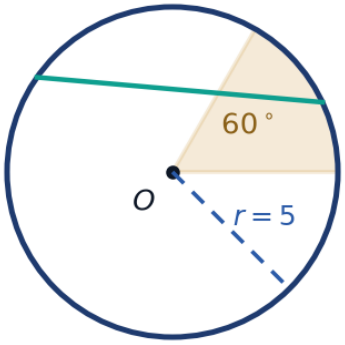
**c.** 17 cm

**d.** 144 cm

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**40** [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?




**a.** 6 units

**b.** $6\sqrt{3}$ units

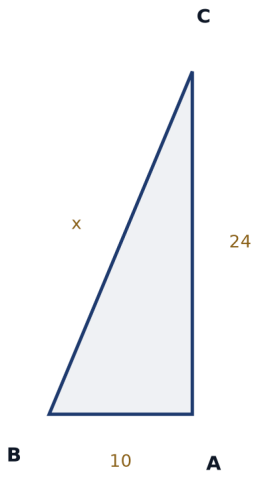
**c.** 12 units

**d.** $12\sqrt{3}$ units

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**41** [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

The diagram shows a right-angled triangle ABC, with the right angle at B. The length of side AB is 10 cm and the length of side BC is 24 cm. A semicircle is drawn with AC as its diameter. Calculate the area of the semicircle in terms of $\pi$.




**a.** $169\pi$ cm$^2$

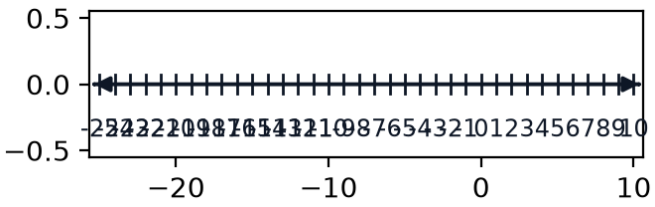
**b.** $84.5\pi$ cm$^2$

**c.** $13\pi$ cm$^2$

**d.** 169 cm$^2$

**42** [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

The diagram shows a number line. The distance between the points marked $A$ and $B$ is $15$ units. The coordinate of $A$ is $-7$. Find the possible coordinates of $B$.




**a.** -22 and 8

**b.** 8 and 22

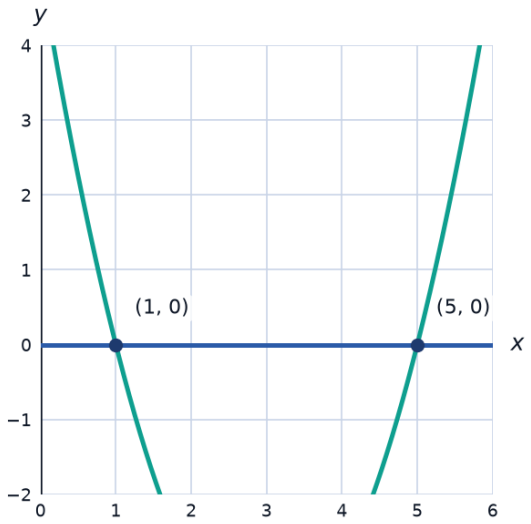
**c.** -22 and -8

**d.** 7 and 22

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**43** [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

The diagram shows the graph of the quadratic function $y = x^2 + bx + c$. The graph intersects the x-axis at $(1, 0)$ and $(5, 0)$. Find the values of $b$ and $c$.




**a.** $b = -6, c = 5$

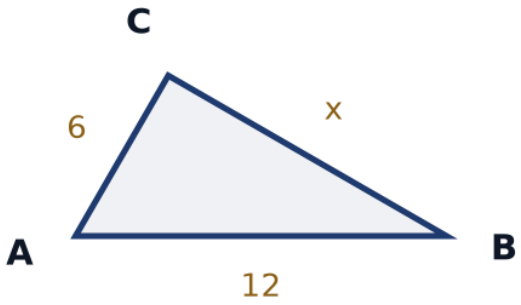
**b.** $b = 6, c = 5$

**c.** $b = -6, c = -5$

**d.** $b = 6, c = -5$

**44** [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 1.01 Functions, Modulus and Quadratics

In a right-angled triangle, one of the acute angles is $30^\circ$ and the side opposite to this angle is $6$ cm. Calculate the length of the hypotenuse.




**a.** 10 cm

**b.** 12 cm

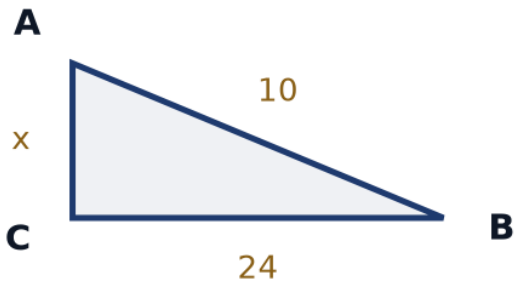
**c.** 8 cm

**d.** 6 cm

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**45** [Challenging] • Target: Permutations & Combinations Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

In a right-angled triangle ABC, where the right angle is at C, if $AC = 10$ and $BC = 24$, find the length of the hypotenuse $AB$.




**a.** 25

**b.** 26

**c.** 27

**d.** 28

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